How do you like your dairy?

A review of how the world tastes milk products

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ABSTRACT

Code available at: https://github.com/dewi-elisa/dairyWorld.

1 INTRODUCTION

Dairy consumption various across cultures. For example, ethnicity affects the liking of yogurt [10]. Moreover, the same dairy products can be perceived as differing in healthiness across cultures [4]. In addition, country characteristics can influence which dairy products are consumed [9]. Therefore, studying the relationship between people and dairy can clarify the broader human relationship with food and deepen the understanding of cultural differences.

Research question. What is the relationship between dairy products and the country of origin?

- For each dairy product, where do they originate from?
- How is the dairy product consumed?
- Is there a relationship between the dairy product and the country of origin?

2 RELATED WORK

Dairy perception and consumption are strongly shaped by culture. At the same time, the history and geography of dairy farming influence how dairy products are produced and valued, and the country of origin of a product can influence consumers. This section goes into more depth on these topics and further discusses food computing and the current research.

2.1 Dairy and culture

A number of studies show that people in different countries perceive and like dairy products in different ways. For example, work on yogurt with different degrees of granularity [10] found that Chinese and Danish consumers responded differently to the presence or absence of particles. Moreover, Johansen et al. [4] studied young Norwegian, Danish, and Californian consumers and showed that all three groups have common motives for consuming calorie-reduced dairy products (e.g. fat content, perceived healthiness and taste), but their relative importance differs across countries. In addition, research on Australian and Chinese consumers [9] reports cross-country differences in the liking of pasteurized milk versus long shelf life milk and in how shelf life labeling affects liking. Lastly, research with a trained panel on flavor lexicons for cheese [1] showed that the panels from New Zealand, Ireland, and the United States used different words to describe the same cheeses, and tended to group cheeses by country of origin.

Beyond sensory perception, cultural and historical factors also influence how dairy is produced, valued, and consumed. For example, in India most milk comes from the buffalo rather than cows because cows have a sacred status in Hinduism. Moreover, in the United States, people are increasingly shifting towards plant-based milks [20]. Lastly, dairy farming can be of great importance to mountain communities [17], family farming cultures [3], and animal welfare [16]. This is further emphasized by Valenze [19], who researched the different religious meanings of milk and the different functions throughout the ages. Thus, dairy is not only a food product, but also has cultural, religious, and socio-economic significance.

2.2 Country-of-origin labeling

Country-of-origin (COO) research has shown that the country where a product is made can influence how consumers evaluate it. This suggests that links between dairy products and countries can shape perceptions and choices. Knight and Calantone [5] propose a model of how COO information is processed cognitively and show that COO effects are complex and cultural context plays an important role in purchase decisions. Moreover, research on the Canadian market [13] found that COO labeling affects the purchase choice and willingness to pay for dairy products. Finally, Yang et al. [22] illustrate how consumer behavior is influenced through COO information after a major food-related scandal.

2.3 Food computing

Food computing and computational gastronomy are emerging research areas that use computational methods to study food-related topics. Surveys and overviews in this area [2, 12] describe how food data (e.g. recipes, images, social media posts) can be analyzed with tools from data science and artificial intelligence to address a wide range of questions about flavor, health, and culinary culture. Several studies focus on culinary cultures and food-country relations. For example, Laufer et al. [8] mine cultural relations between different language communities in Europe by analyzing how food cultures are described on Wikipedia pages across different language editions. Moreover, palette varieties around the world are investigated [15] and are found to have strong geographical and cultural similarities between countries. Furthermore, a large visual question answering dataset with text-image pairs is made identifying dish names and their origins [21]. Lastly, two topic modelling approaches are applied to recipes [7, 11] to discover more about the human relationship with food.

2.4 Current research

Previous research has shown that culture influences how dairy products are perceived and consumed, and that the country of origin of a product can affect how consumers evaluate it. Moreover, work on food computing has demonstrated that large-scale online data and computational methods can be used to study food-related topics, including culinary cultures and food-country relations. However,

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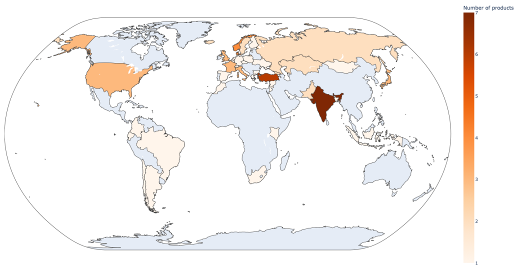

Figure 1: Origins of dairy products

there is little work that combines these perspectives to study how dairy products and countries are linked in textual descriptions. Therefore, this paper uses text mining on Wikipedia articles to investigate how dairy products are associated with their countries of origin.

3 APPROACH

3.1 Data

For this project, Wikipedia pages of dairy products and countries are used. All pages are linked in a Wikipedia page with a list of dairy products[1]. Only the products with all information present (product name, product URL, origin(s), origin URL(s)) are used, resulting in a total of 60 dairy products out of 130. All data is scraped on 5 January 2026 using the BeautifulSoup library in Python[2] and further processed using pandas [14].

3.2 Method

The linked Wikipedia page provides a list of dairy product and origins, answering the first subquestion.

3.3 Validation

For the first subquestion, the results are validated by randomly picking 6 dairy products and checking their origins with other sources.

4 RESULTS

4.1 Product origins

Figure 1 shows the origins of the dairy products investigated in this research. The darker the orange, the more products originate from a country. A more detailed and interactive version of this figure can be found at Github[3]. The interactive version shows on hover which specific products originated from that country. Appendix B describes how this figure was made.

[1]https://en.wikipedia.org/wiki/List_of_dairy_products
[2]https://pypi.org/project/beautifulsoup4/
[3]https://dewi-elisa.github.io/dairyWorld/results/world_map.html

4.2 Consummation methods

4.3 Product-country relations

4.4 Validation

5 DISCUSSION

5.1 Validation

* What the validation implies for trustworthiness * How it affects interpretation of your main findings (do results hold, where to be cautious).

5.2 Limitations

* not a lot of products (only 60 out of 130) * only Wikipedia data, with ambiguous regional terms like "Scandinavia" * bias toward well-documented countries

5.3 Future work

* more data sources * more products * more in-depth analysis -> maybe different, data-driven, descriptors instead of hand-picked?

6 CONCLUSION

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A WORK REPORT

In order to find literature, I mainly looked in an online university library. I used queries such as 'milk', 'dairy', 'country of origin dairy', and 'food computing'.

During the programming, I mainly looked at documentation and examples online on websites such as https://www.geeksforgeeks.org and https://www.w3schools.com. One problem I encountered was the Robot Policy of Wikipedia, https://wikitech.wikimedia.org/wiki/Robot_policy#REST_API_rules, which prevented me from scraping the Wikipedia pages. I solved this by adding a user-agent to my requests.

B WORLD MAP

In order to be able to make Figure 1, ISO3 mapping was needed for each country of origin. The country converter package [18] was used for this task. However, not all countries were countries: some were cities, regions, former countries, or cultures. Therefore, a

Country	ISO3
USSR	Russia
Khan garh	Pakistan
Scotland	United Kingdom
Vikings	United Kingdom
Central Asia	Kazakhstan
West Sumatra	Indonesia
Scandinavia	Sweden, Norway, Denmark
Alps	Switzerland
Andhra Pradesh	India
Caucasus	Georgia
Punjab	India, Pakistan
Central and Eastern Europe	Poland
Mannheim	Germany
Minoru Shirota	Japan

Table 1: Manual mapping of non-country origins to countries for ISO3 conversion

manual mapping was made for these cases to a country. The manual mapping to that country is shown in Table 1. For the cities and regions, the country where they are located was chosen. For the former countries, the country that is most closely related to them was chosen. For the Viking culture, the United Kingdom was chosen as the Wikipedia page with dairy product mentions Scotland. When a part of a continent was mentioned, a country central in that part was chosen. Together with product and country names, this was then put in a choropleth map using Plotly [6].